

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper No. BPT-502: Classical Mechanics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **four** of the following:

[4 × 5 = 20]

- (a) How many generalized/quasi-generalized coordinates are required to describe the following systems?
- (i) A rigid body in 3 dimensions.
 - (ii) A solid sphere rolling without slipping on an inclined plane.
 - (iii) A spherical pendulum in 3 dimensions.
- (b) What is isotropy of space? Prove that the conservation law of angular momentum follows from it.
- (c) A particle moving in a central force field located at $r = 0$ describes a spiral orbit $r = a\theta$. Find the force law.
- (d) The Lagrangian of a system is given by:

$$L = \frac{1}{2}m\dot{\vec{r}}^2 + \dot{\vec{r}} \cdot (\vec{a} \times \vec{r}) + V(\vec{r})$$

where $\vec{a}$ is an arbitrary vector depending on time only. Find the equations of motion for the system.

- (e) A particle is constrained to move on the surface of a cylinder $x^2 + y^2 = R^2$ and is subjected to a force directed towards the origin and proportional to the distance of the particle from the origin. Find the equations of motion for the particle using the Lagrange multiplier method.

Question 2.

- (a) Consider a particle of mass m free to slide on a smooth helical wire whose equations in cylindrical coordinates are given by $\rho = R, z = k\theta$.
- (i) Construct the Lagrangian of the system. [4]
 - (ii) Find the equation of motion and solve it to discuss the motion of the particle (if it is released at rest at $\rho = R, z = z_0, \theta = \theta_0$). [5]
 - (iii) Construct the Hamiltonian of the system and find the canonical equations of motion. [5]

OR

- (a) Show that the Lorentz force on a charged particle in an electromagnetic field can be obtained from a velocity-dependent potential $U = q\phi - \frac{q}{c}\vec{v} \cdot \vec{A}$. [5]
- (b) Write the Lagrangian of a charged particle in the electromagnetic field using this potential. [4]
- (c) Write down the Hamiltonian for a system consisting of an electron, a proton, and a neutron. How will the Hamiltonian be modified in the presence of a uniform magnetic field $\vec{B} = B_0(\hat{i} + 2\hat{j})$? [5]

Question 3.

- (a) A particle of mass m is moving on a plane under an attractive central force given by:

$$F(r) = -\frac{\alpha}{r^2} - \frac{\beta}{r^3}, \quad \alpha, \beta > 0$$

- (i) Find the effective potential for this system. [3]
- (ii) Discuss the motion of the particle qualitatively for various total energy values of the particle. [4]
- (iii) Show that the angular momentum of the particle is conserved. [3]
- (iv) Show that the position vector of the particle sweeps out equal areas in equal times. [4]

Question 4.

- (a) Define canonical transformation. How is it useful? [4]
- (b) For what values of α and β is the transformation:

$$Q = q^\alpha \cos(\beta p), \quad P = q^\alpha \sin(\beta p)$$

canonical? [4]

- (c) Find a suitable generating function for this canonical transformation. [6]

OR

- (a) Express the Hamiltonian $H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2$ in terms of Q, P by using a canonical transformation generated by $F_1(q, Q) = \frac{1}{2}m\omega q^2 \cot Q$. [7]
- (b) Calculate the Poisson brackets:

$$\{L_x, x\}, \quad \{L_x, p_y\}, \quad \text{and} \quad \{(\vec{r} \times \vec{p})_z, z\}$$

[7]

Question 5.

- (a) Show that if A and B are two constants of motion, then their Poisson bracket $\{A, B\}$ is also a constant of motion. [6]
- (b) State and prove the virial theorem. [8]

OR

- (a) A particle of mass m is falling freely from a height h . Find the solution for the particle using the Hamilton-Jacobi equation. [7]
- (b) Discuss the idea of mechanical similarity. Use this to show that the time of fall for a freely falling body is proportional to the square root of its initial height. [7]